

Bryan Djenabia Joseph

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EDUCATION

San Diego State University

B.S. in Computer Science | GPA: 3.85 / 4.0

Relevant Coursework: Data Structures, Computer Organization & Assembly, Discrete Mathematics, Algorithms.

San Diego, CA

Expected May 2028

TECHNICAL SKILLS

Languages: Java, Python, C++, Swift, TypeScript, JavaScript, SQL, Bash, HTML/CSS

Frameworks & Libraries: Spring Boot, React, SwiftUI, Combine, Node.js, Pandas, NumPy, Matplotlib, JUnit, Vitest

Systems & Infrastructure: Concurrency & multithreading, event-driven architecture, REST APIs, WebSockets, Docker, CI/CD (GitHub Actions), Linux, TCP/IP, caching, observability

Data & Tools: PostgreSQL, SQLite, Supabase, Redis, Git, Gradle, Vite, OpenAI APIs, Stripe, Monte Carlo simulation, backtesting & statistical analysis

EXPERIENCE

Discovery Partners Institute (University of Illinois System) - Chicago, IL

Selected three times for the competitive Digital Scholars program:

- Student Intern (Jun 2023 - Aug 2023): data science track—data preparation, visualization, and machine learning on real projects.
- Youth Mentor (Jan 2023 - May 2023): mentored a cohort of 25 middle-school students from zero experience to a working iOS app; 100+ students taught across all cohorts.
- Summer Intern (Jun 2022 - Jul 2022): first exposure to iOS development - Swift fundamentals, building user-facing apps end-to-end.

PROJECTS

StayFit – AI Health & Fitness | [Live on the App Store](#)

Swift, SwiftUI, Supabase, Stripe

- Designed and shipped a **native iOS app live on the App Store** — 80,000+ lines of Swift across 260+ files (58 services, MVVM + Combine) spanning workout and nutrition tracking, a social feed, a creator marketplace, an Apple Watch app, and home-screen widgets.
- Built a real-time exercise **form-check engine on Apple's Vision framework**, scoring live body-pose and joint angles on-device to give users instant coaching feedback.
- Engineered an **offline-first sync layer** (persisted mutation queue, last-write-wins conflict resolution) backed by 12 Supabase/PostgreSQL edge functions powering Stripe Connect payouts, AI chat, and referral rewards.

WebMARS – Browser-Based MIPS Assembler & Simulator | [webmarsimulator.com](#)

TypeScript, React, Vite

- Led core-engine development of a **full MIPS32 toolchain running entirely in-browser** — lexer, two-pass assembler (~50 instructions plus pseudo-instructions), FPU, and traps — in 15,000+ lines of strict TypeScript, deployed live to production.
- Architected a clean core/UI split (pure-TypeScript engine behind a Zustand store) that enabled a **time-travel debugger** rewinding register, FPU, and memory state through a circular snapshot history.
- Wrote **100+ unit and integration tests (Vitest)** gating every pull request via GitHub Actions CI, catching sign-extension and address-resolution edge cases across the instruction set.

Futures Trading Engine & Risk Dashboard

Java 21, Spring Boot, React/TS

- Built an **event-driven, multithreaded trading engine in Java 21** with a bounded concurrent event bus and decoupled data, strategy, risk, and execution modules processing high-frequency tick streams across 8 futures instruments.
- Implemented a **risk pipeline enforcing daily-loss, trailing-drawdown, and contract limits on 100% of orders**, with automated flatten-by-time and kill-switch logic guaranteeing rule compliance.
- Developed a **Monte Carlo simulator (10,000+ paths, block bootstrapping)** and a historical backtesting harness measuring win rate, R-multiples, and drawdown, streaming live PnL to a Spring Boot + React dashboard over WebSockets.

LEADERSHIP & ACHIEVEMENTS

- **James Dyson Foundation Scholar (2024)** • **Google Code Summit Hackathon Winner — Chicago, IL (2024)**
- Member, National Society of Black Engineers (NSBE) and Association for Computing Machinery (ACM), SDSU — Nov 2024 – Present.
- Languages: French (Fluent), Spanish (Advanced — California Seal of Biliteracy).